LeetCode

Платформа за упражнение по алгоритми и структури от данни чрез интерактивни задачи, тестове и състезания.

1. **Резултати от създадените функции, приложени върху данните от база данни LeetCodeDb:**

- **Функция RankUserAmongAll, която определя мястото на потребителя сред всички по общ брой успешни решения;**

```
create function dbo.RankUserAmongAll (@UserID int)
returns int as
begin
   declare @rank int;
   with Ranking as (select  u.UserID,
           row_number() over (order by coint(case when s.IsAccepted = 1 then 1 end)
desc) as Rank
       from [User] u
       left join Submission s on u.UserID = s.UserID
       group by u.UserID)
   select @rank = Rank
   from Ranking
   where UserID = @UserID;
   return isnull(@rank, 1);
end;
select u.UserID, u.Username, dbo.RankUserAmongAll(2) as Rank
        from [User] u where u.UserID = 2;
```

Results | Messages

	UserID	Username	Rank
1	2	tania	12

	UserID	Username	SolvedProblems	Rank
3	7	marin	2	3
4	9	yang	2	4
5	10	maria	2	5
6	12	lidia	1	6
7	13	diana	1	7
8	14	valentin	1	8
9	15	stefan	1	9
10	8	robin	1	10
11	1	marko	1	11
12	2	tania	1	12
13	3	igor	1	13
14	4	sergey	1	14

Фигура 1. Резултат от функцията RankUserAmongAll.

- **Функция GetPredictedDifficulty, имитация на “machine-learning–подобен” анализ, който:**
  - **отчита исторически успех,**
  - **сложност на задачите,**
  - **процент приети решения.**

```
create function dbo.GetPredictedDifficulty (@UserID int)
return nvarchar(20) as
```

```
begin
   declare
      @AvgDifficulty float, @ErrorRate float, @AvgRuntime float, @Prediction
nvarchar(20);
   select @AvgDifficulty = avg(
      case p.Difficulty
         when 'Easy' then 1
         when 'Medium' then 2
         when 'Hard' then 3
      end)
   from Submission s
   join Problem p on s.ProblemID = p.ProblemID
   where s.UserID = @UserID and s.IsAccepted = 1;
   if @AvgDifficulty is null
      return 'Easy';
   select @ErrorRate =
      cast(sum(case when IsAccepted = 0 then 1 else 0 end) as float) / count(*)
   from Submission
   where UserID = @UserID;
   select @AvgRuntime = avg(Runtime)
   from Submission
   where UserID = @UserID
    and IsAccepted = 1;
   if (@AvgDifficulty >= 2.5 and @ErrorRate < 0.20)
      set @Prediction = 'Hard';
   else if (@AvgDifficulty >= 1.8 and @ErrorRate < 0.45)
      set @Prediction = 'Medium';
   else
      set @Prediction = 'Easy';
   return @Prediction;
end;
select u.UserID, u.Username, dbo.GetPredictedDifficulty(u.UserID) as
RecommendedDifficulty from [User] u;
```

100 %

Results | Messages

	UserID	Username	RecommendedDifficulty
1	1	marko	Easy
2	2	tania	Easy
3	3	igor	Easy
4	4	sergey	Easy
5	5	katya	Hard
6	6	lev	Easy
7	7	marin	Easy
8	8	robin	Easy
9	9	yang	Easy
10	10	maria	Medium
11	11	tomas	Easy
12	12	lidia	Medium
13	13	diana	Easy
14	14	valentin	Easy
15	15	stefan	Easy

Фигура 2. Резултат от функцията GetPredictedDifficulty.

2. **Резултати от създадените съхранени процедури, приложени върху данните от база данни LeetCodeDb:**
   - **Съхранена процедура GetProblemsByDifficulty - филтър по трудност;**

```
create procedure GetProblemsByDifficulty
   @Difficulty nvarchar(50)
as begin
   select ProblemID, Title, Difficulty, CreatedAt
   from Problem where Difficulty = @Difficulty;
end;
exec GetProblemsByDifficulty @Difficulty = 'Hard';
```

Results | Messages

	ProblemID	Title	Difficulty	CreatedAt
1	5	Edit Distance	Hard	2025-11-22 14:35:04.520
2	6	Longest Increasing Path	Hard	2025-11-22 14:35:04.520
3	11	Daily Revenue Moving Average	Hard	2025-11-22 14:35:04.520
4	15	Dijkstra Shortest Path	Hard	2025-11-22 14:35:04.520

Фигура 3. Резултат от съхранената процедура GetProblemsByDifficulty.

   - **Съхранена процедура AddSubmission - добавяне на решение**

```
create procedure AddSubmission
   @UserID int, @ProblemID int, @LanguageID int, @Code
nvarchar(max)
as begin
   insert into Submission (UserID, ProblemID, LanguageID, Code,
Result, IsAccepted)
   values (@UserID, @ProblemID, @LanguageID, @Code, 'Pending',
0);
end;
exec AddSubmission
   @UserID = 1, @ProblemID = 2, @LanguageID = 1, @Code =
N'print("Hello")';
select * from Submission order by SublissionID desc;)
```

Results | Messages

	SublissionID	UserID	ProblemID	LanguageID	Code	Result	Runtime	MemoryUsage	IsAccepted	SubmittedAt
1	31	1	2	1	print("Hello")	Pending	NULL	NULL	0	2025-11-24 11:41:27.003
2	30	1	2	1	print("Hello")	Pending	NULL	NULL	0	2025-11-22 19:19:37.770
3	29	10	20	3	code...	Accepted	41	5.1	1	2025-03-13 11:02:00.000
4	28	9	19	1	code...	Accepted	81.3	6.2	1	2025-03-13 07:45:00.000
5	27	8	18	3	code...	Runtime Error	0	0	0	2025-03-12 18:00:00.000
6	26	7	17	1	code...	Accepted	2.8	0.9	1	2025-03-12 18:11:00.000

Фигура 4. Резултат от съхранената процедура AddSubmission.

3. **Резултати от създадените тригери, приложени върху данните от база данни LeetCodeDb:**
   - **Тригер trg_Discussion_AddReply.**

```
create trigger trg_Discussion_AddReply
on Discussion
after insert
as begin
   update Discussion
   set RepliesCount = isnull(RepliesCount, 0) + 1
```

```
    where DiscussionID in (select DiscussionID from inserted);
end;

insert into Discussion (ProblemID, AuthorID, Message) values (2, 1, N'New test reply');

        select DiscussionID, ProblemID, AuthorID, Message, RepliesCount from Discussion order by DiscussionID desc;
```

	DiscussionID	ProblemID	AuthorID	Message	RepliesCount
1	11	2	1	New test reply	1
2	10	13	12	Dijkstra + priority queue works great.	0
3	9	10	6	Good classical question.	0
4	8	9	11	Window functions are fun.	5
5	7	7	10	SQL JOINs are not that scary.	11
6	6	5	8	Took me 2 hours but finally got it.	2
7	5	4	5	Nice DP problem!	0
8	4	3	2	Kadane is the cleanest solution.	1
9	3	2	7	Sliding window helped a lot.	0
10	2	1	4	Two pointers also works but needs sorting.	4
11	1	1	3	I solved this using a hash map in O(n).	0

Фигура 5. Данни в таблицата Discussion.

- **Тригер trg_Problem_Update.**

```
create trigger trg_Problem_Update
on Problem
after update
as begin
    update Problem
    set UpdatedAt = getdate()
    where ProblemID in (select ProblemID from inserted);
end;

update Problem
set Title = 'Modified problem title'
where ProblemID = 2;

select ProblemID, UpdatedAt from Problem where ProblemID = 2;
```

	ProblemID	Title	UpdatedAt
1	2	Modified problem title	2025-11-22 19:27:44.170

Фигура 6. Данни в таблицата trg_Problem_Update.